

Service Oriented Architecture (SOA)

SOA

- Understand the SOA architecture

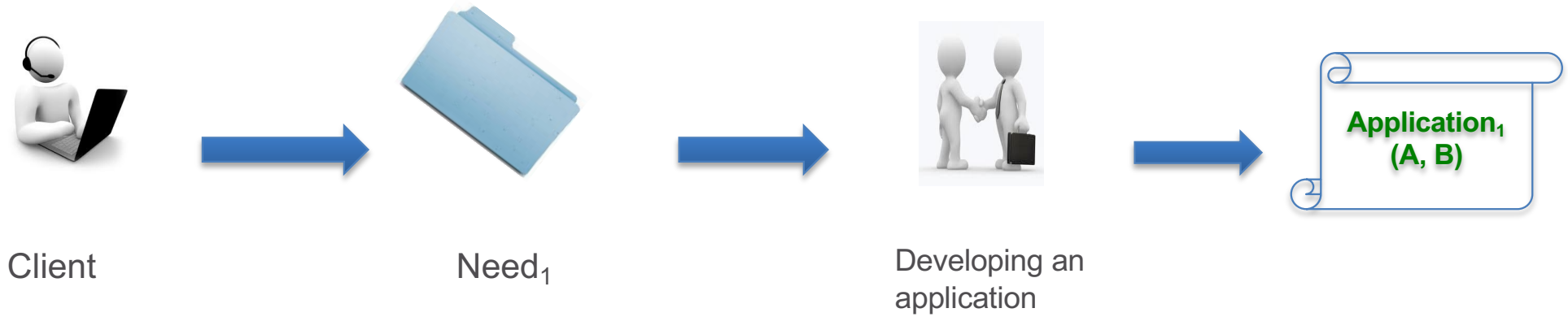
SOA

- Understand the SOA architecture
- Understand Web services

SOA

- Understand the service oriented architecture
- Understand Web services
- Understand the underlying issues of the development of distributed applications

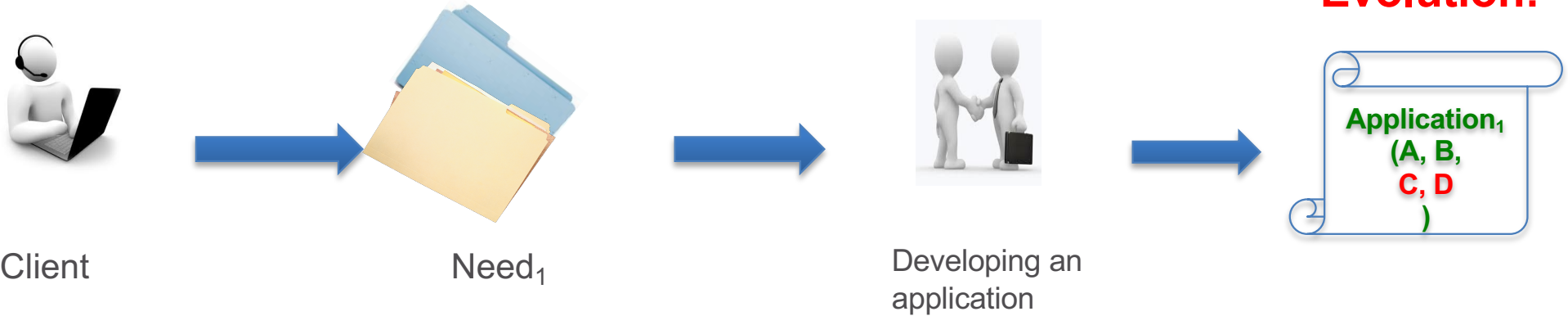
Evolution of applications



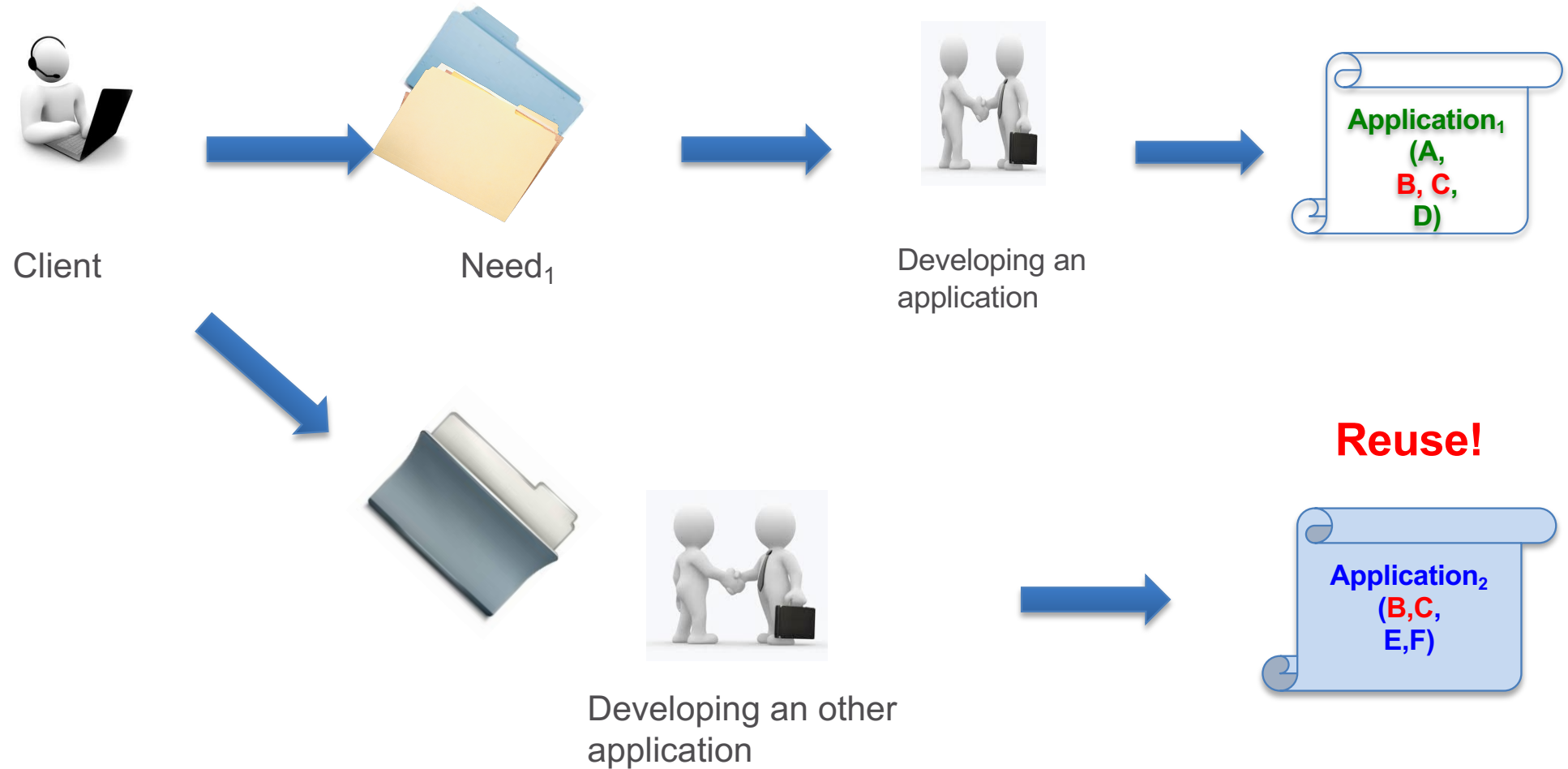
Evolution of applications

INSA

Evolution!



Applications reuse



Distributed applications

- Needs are usually complex
- It is necessary to build inter-organizational collaborations



Client

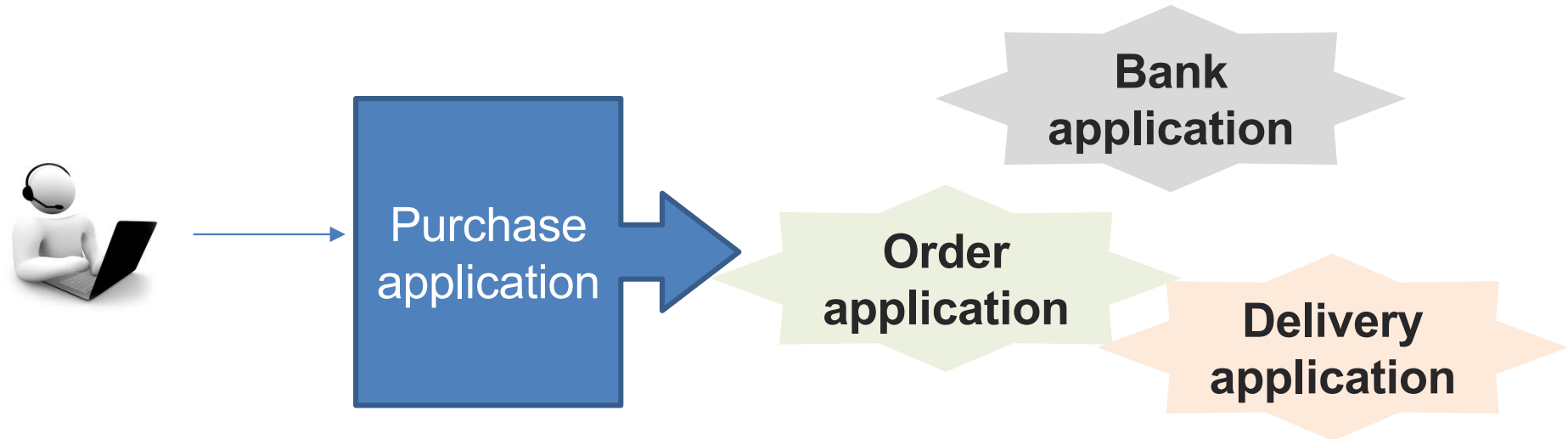


Complex needs



Integration of different inter-organizational applications

Distributed applications



Application integration

- Applications can be heterogeneous
- Different programming languages
- Different execution platforms

Application integration

- Applications can be heterogeneous
- Different programming languages
- Different execution platforms
- **Application integration** problem!

Middleware for the application integration

- DCOM
- CORBA
- EJB
- RMI

Middleware for the application integration

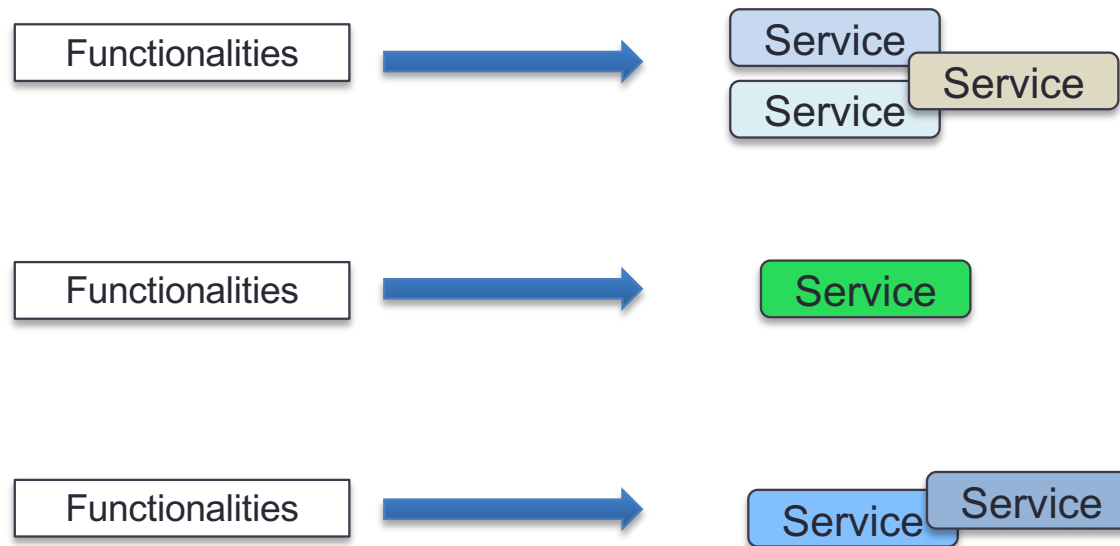
- Middleware-based solution are limited !
- Interoperability is possible if the components use the same solution
- Specific transport protocols require a specific network configuration

Service Oriented Architecture - SOA

SOA ?

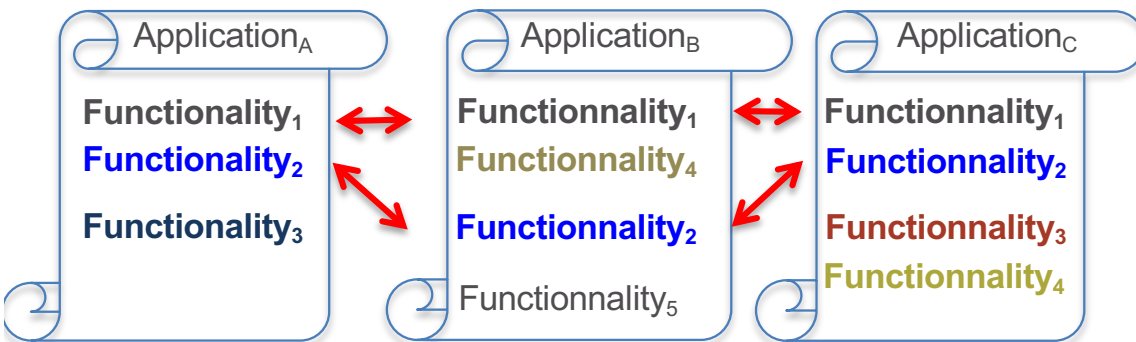
SOA

- Encapsulation of business applications as services

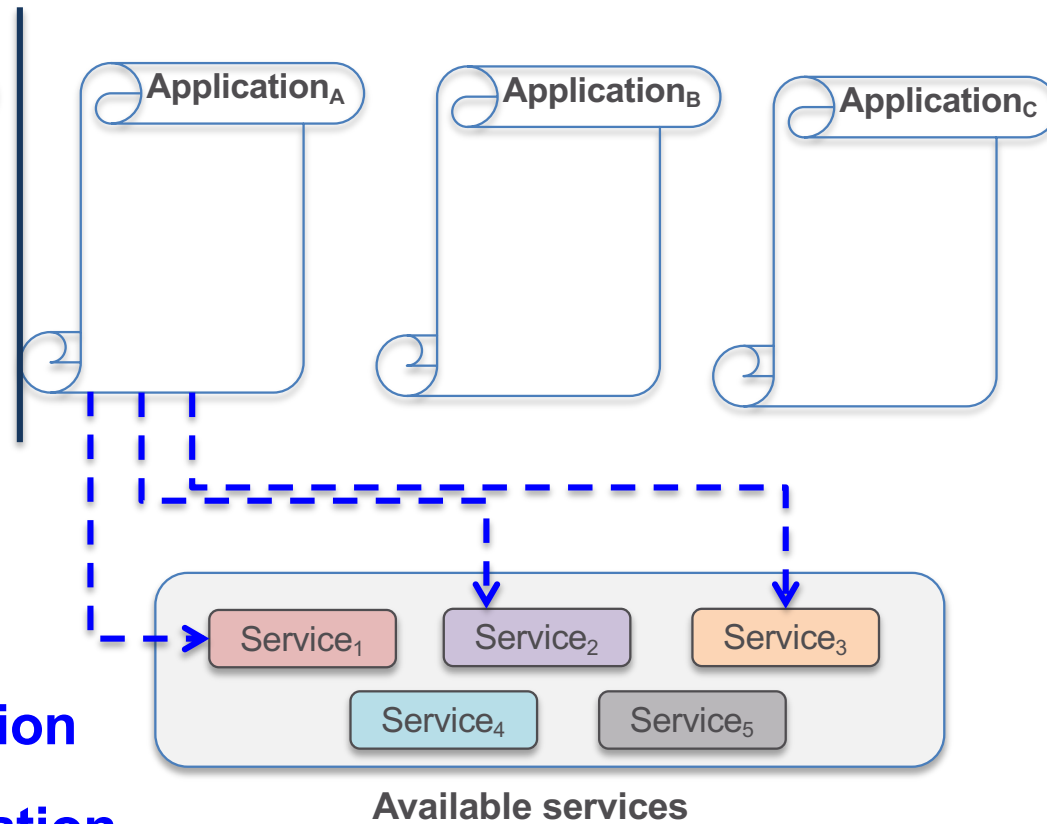


SOA

Without SOA



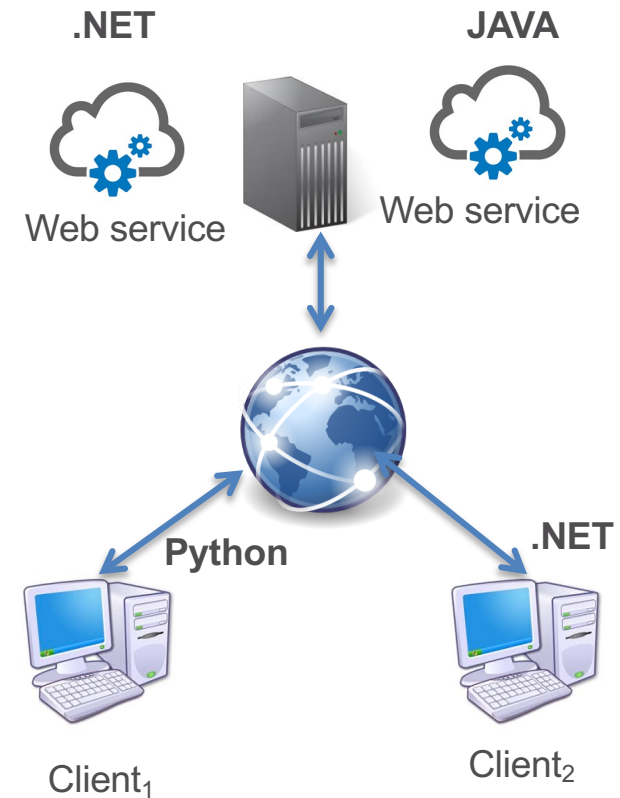
With SOA



- ✓ Service reuse
- ✓ Application evolution
- ✓ Application integration

Web service

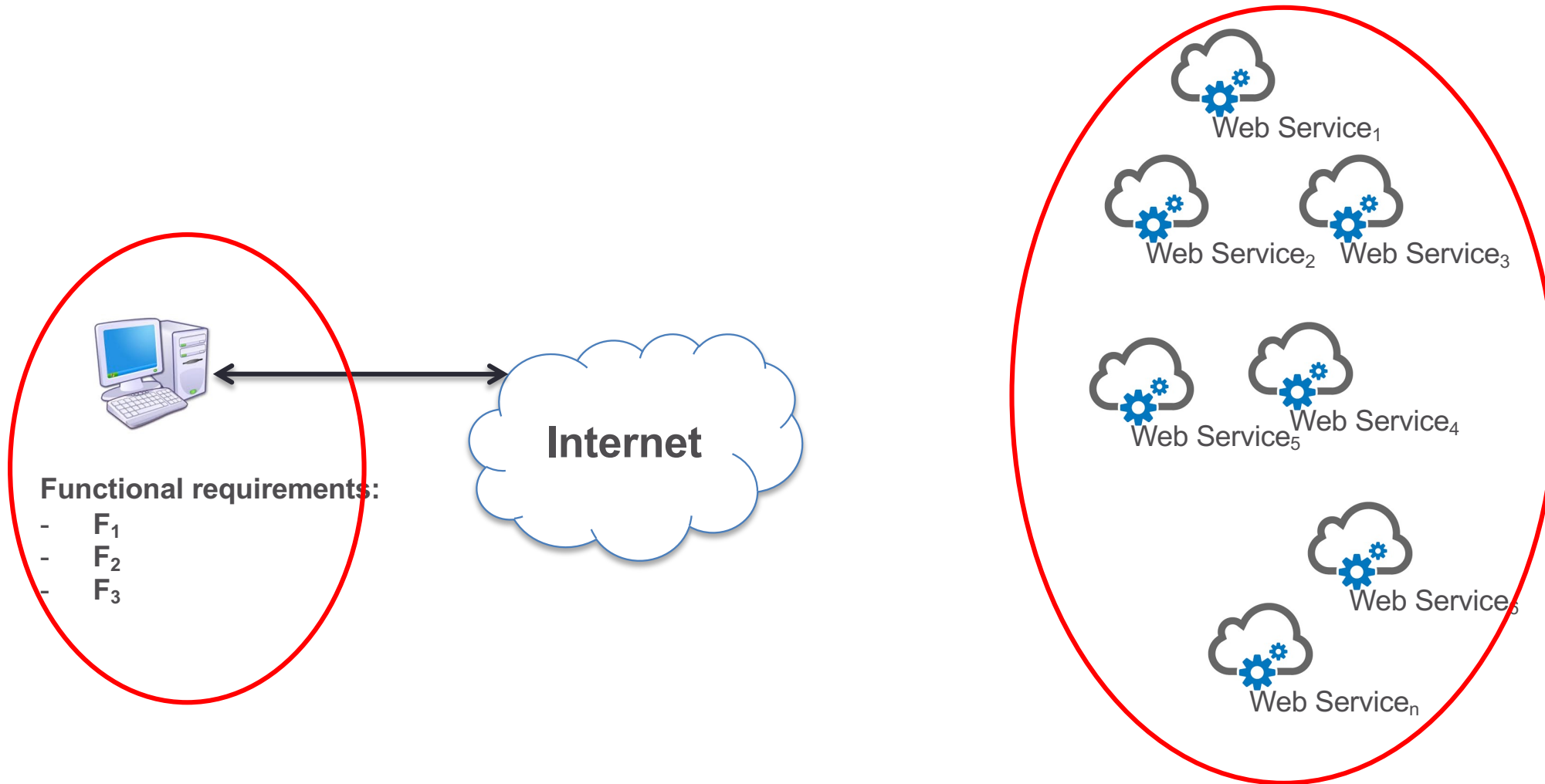
- A software accessible through **standard protocols of the Web**
- Interface that provides data and services
- Decorrelation with platforms and programming languages



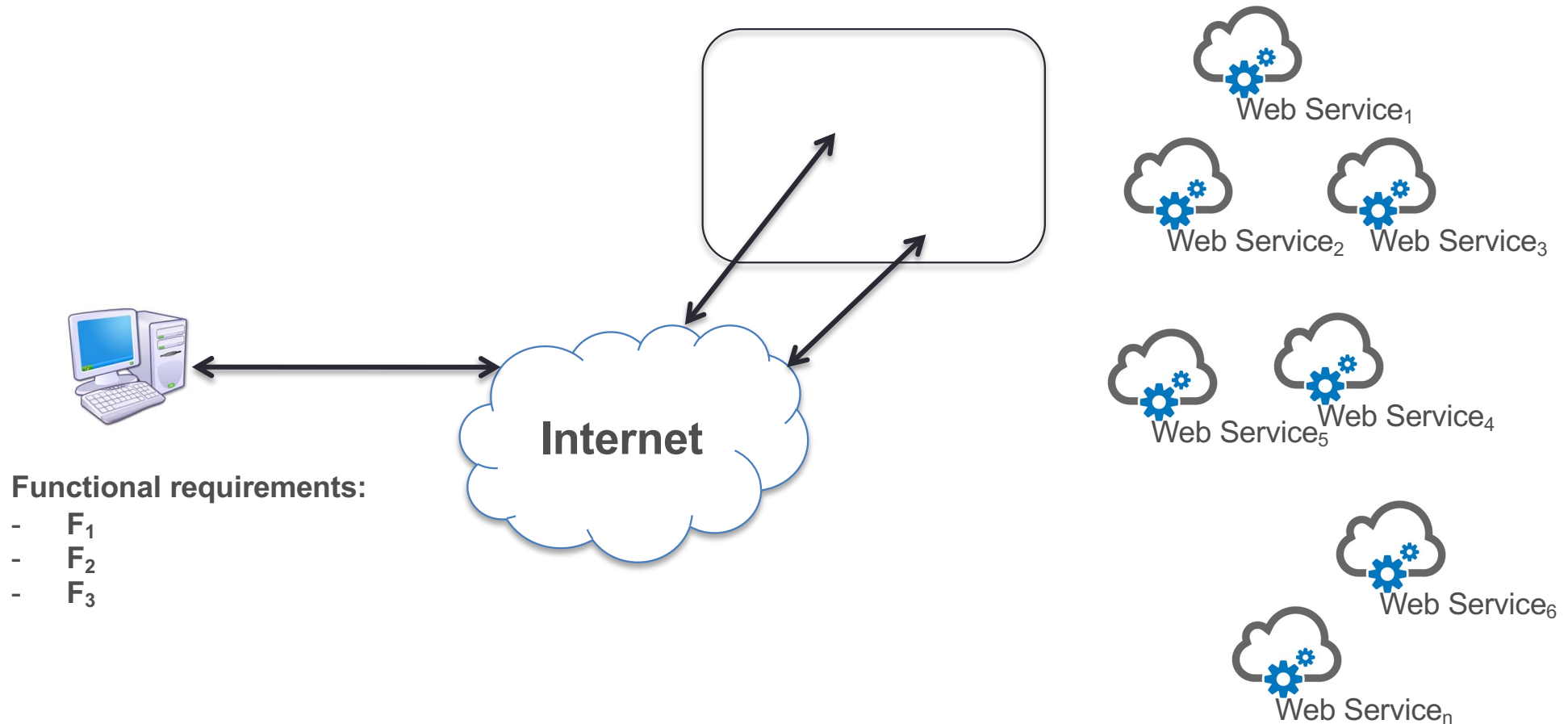
Web services properties

Web services discovery

Web service discovery

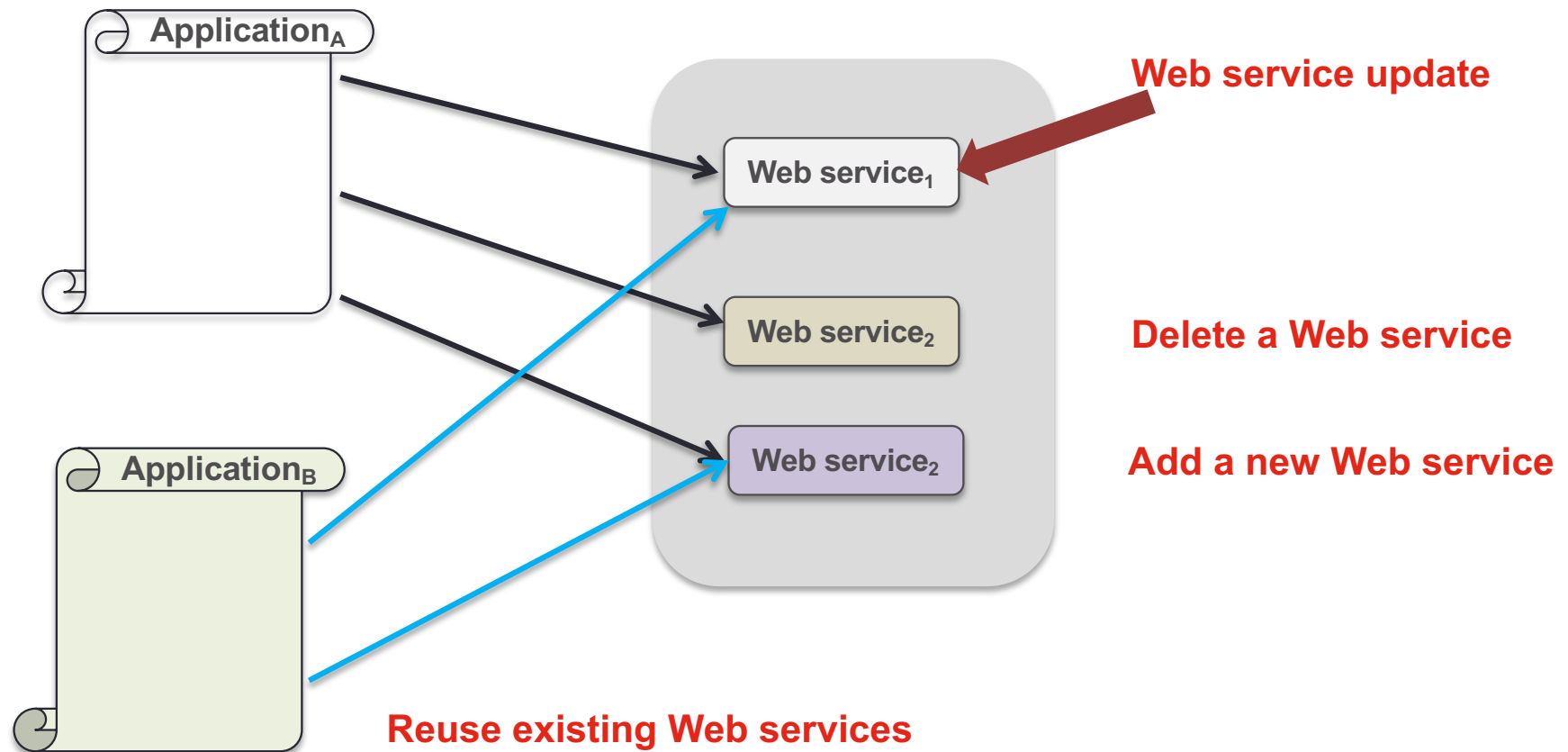


Web service discovery



Flexibility, maintainability, and reuse

Flexibility, maintainability, and reuse



Web service composition

Web service composition

- Provides new value added Web services

Different languages

